

Teaching a Fracture Classifier Where to Look

CNN vs ScatNet, six XAI methods and a joint classifier–detector on FracAtlas

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Data: FracAtlas (Abdeen et al., *Scientific Data*, 2023)
Code: github.com/silvergjeka22/bone-fracture-detection

Outline

Part I – Classify and explain

1. The problem and the data
2. CNN vs ScatNet (+ ResNet18)
3. Six XAI methods: how they work
4. Faithful? On the fracture?

Part II – Teach it where to look

5. A first attempt that failed
6. Our joint model
7. YOLO proposes, our model decides
8. The pipeline, limits and conclusion

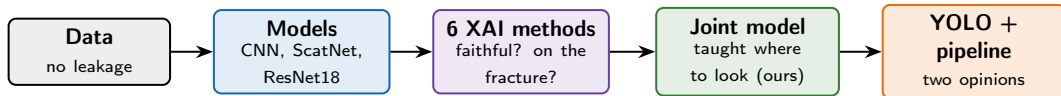
Part I

Classify and Explain

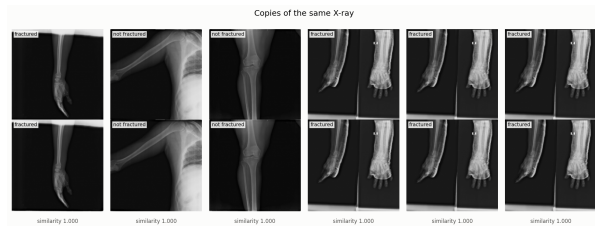
Which model sees fractures, and where does it look?

The Problem: Right, but for the Right Reason?

- A model can be **accurate for the wrong reasons**: a metal plate, a cast or the body part can predict “fractured” better than a thin fracture line.
- **XAI** draws a heatmap of the evidence. It helps only if it is **faithful** (shows what the model uses) **and** the model uses the **fracture**.
- FracAtlas has the **radiologists’ boxes**: we can **measure** where a model looks.



The Data: FracAtlas, Without Leakage



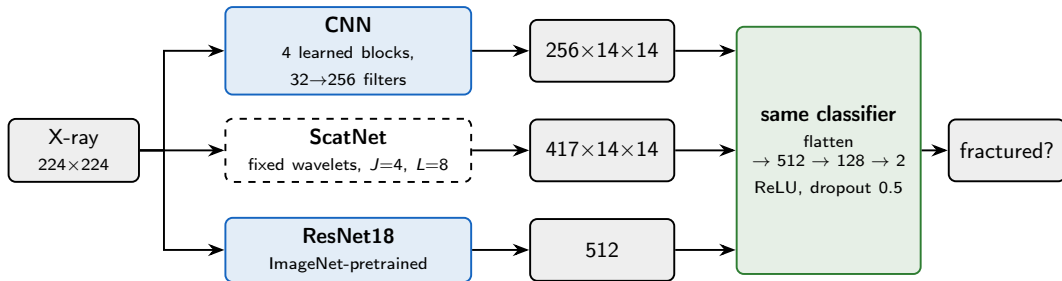
Copies of the same X-ray (flipped, resized, brighter) found automatically.

- 4,083 X-rays; **717 fractured (1 in 6)**, 922 boxes.
- Grey, 224×224 (448 in our method), values in $[-1, 1]$.
- **8.6%** of the X-rays have a copy (158 groups): a split **keeps every group together**.

	X-rays	fractured	boxes
train	2,920	511	651
val	577	102	136
test	586	104	135

Always saying “not fractured” already scores **82.3%**: look at **F1** and **fractures found**.

Three Feature Extractors, One Classifier



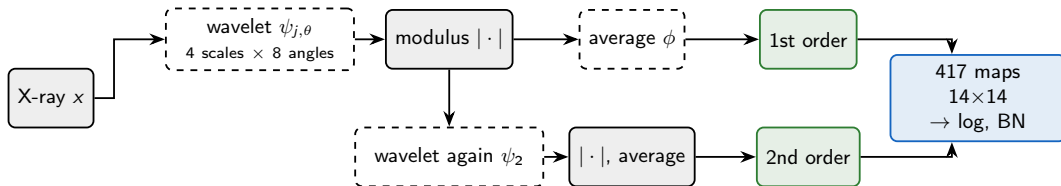
CNN: every filter **learned** from our 2,920 X-rays (first 7×7).

ScatNet: **nothing learned** before the classifier; stable to small deformations.

ResNet18: filters learned on **1.2M ImageNet** photos, then fine-tuned.

Only the classifier's **input size** differs (50,176 / 81,732 / 512). Weighted loss (1 in 6 fractured), 20 epochs, 5-fold grouped CV.

How ScatNet Works



The idea (Bruna & Mallat, 2013)

- A CNN with **fixed** filters: Morlet wavelets at every scale and angle.
- Modulus + averaging $\Rightarrow$ **stable to small shifts and deformations**.
- $1 + 32 + 384 = 417$ coefficient maps (Kymatio, $J=4$, $L=8$).

For fractures

- + No filters to learn: good with **little data**.
- – A small break **is** a small deformation: the averaging blurs exactly what we need.
- – Every coefficient is kept: the classifier has 41.9M parameters.

CNN vs ScatNet (+ ResNet18): Results

	CNN	ScatNet	ResNet18
CV accuracy	73.0±9.0	77.5±2.9	89.0±1.0
CV F1	35.7±13.0	38.0±2.6	66.6±2.7
test accuracy	80.9	80.0	91.5
test F1	23.3	31.6	73.7
fractures found	16%	26%	67%
AUC	68.6	72.7	91.2
parameters	26.1M	41.9M	11.5M

McNemar: CNN $\approx$ ScatNet ($p = 0.55$); ResNet18 beats both ($p < 0.001$).

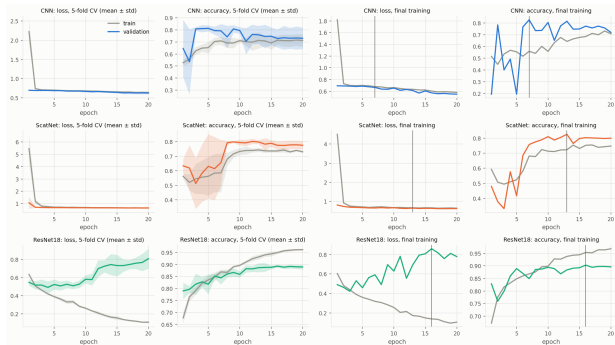
Why CNN and ScatNet fail

- Both **underfit** (loss stuck at 0.6) and stay **below** the 82.3% of “never fractured”.
- Only 511 fractured X-rays; a fracture is 1–2 px wide at 224 px.
- Flattening a 14×14 grid: 26–42M parameters in the classifier.

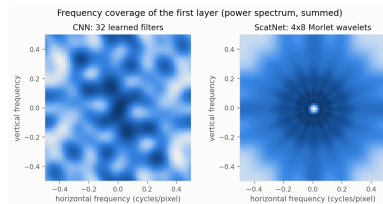
Why ResNet18 wins

- Pretrained edge and texture detectors that 2,920 X-rays cannot teach.
- It overfits after ≈ 10 epochs: the best val epoch is kept.

Learning Curves and Filters



Train (grey) and validation (colour): 5-fold CV and final training.



First-layer filters in frequency.

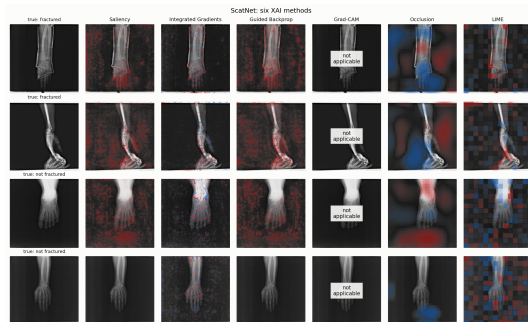
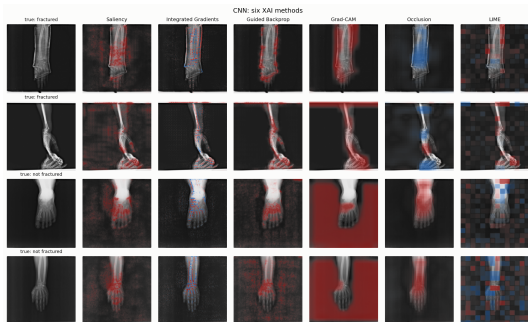
- CNN: noisy learned filters, **uneven** coverage.
- ScatNet: clean wavelets, **every** scale and angle.
- Clean filters are not enough: ScatNet lacks a **learned** hierarchy above them.

Six XAI Methods: How Each One Works

	how it works	strength / weakness	s/X-ray
Saliency	$ \partial z_c / \partial x $: score change for a tiny change of each pixel	fast / noisy, saturates	0.015
Integrated Gradients	gradients averaged on the path black $\rightarrow$ x, times $(x - x')$; sums to $z_c(x) - z_c(x')$	complete / slower	0.194
Guided Backprop	gradient, negative signals stopped at every ReLU	sharp / like an edge detector	0.010
Grad-CAM	last conv layer: channels weighted by their mean gradient, ReLU	class-specific / coarse; not on ScatNet	0.005
Occlusion	slide a black 16×16 window, record the score drop	honest / slow	0.767
LIME	switch random 16×16 patches off, fit a ridge regression	honest / random, slowest	1.377

Gradient methods ask “what if this pixel changed a little?”; **perturbation** methods (Occlusion, LIME) really **remove** parts of the X-ray (black = removed). Grad-CAM needs a learned convolutional layer, which ScatNet does not have.

The Six Methods on the Same X-rays



CNN: 2 fractured, 2 healthy (red = for, blue = against).

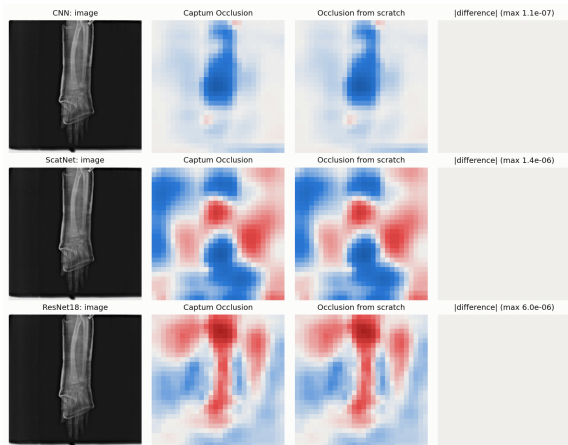
ScatNet: Grad-CAM not applicable.

- The six maps are **visibly different**: there is no single “true” explanation.
- CNN’s Grad-CAM covers large areas; ScatNet’s maps are spread all over the image.

Occlusion, Written From Scratch

Given: model f , X-ray x , class c , window $w=16$,
stride $s=8$
 $z \leftarrow f(x)_c$; $A \leftarrow 0$; $N \leftarrow 0$
for every window position (i, j) , in batches **do**
 $\hat{x} \leftarrow x$ with the window set to black
 $A[\text{window}] += z - f(\hat{x})_c$
 $N[\text{window}] += 1$
end for
return A/N $\triangleright$ mean drop per pixel

- Plain PyTorch, all positions scored in batches.
- Same answer as Captum on all three models: Pearson $r = 1.000$, largest difference $8.6 \cdot 10^{-6}$.



Our Occlusion, Captum's, and their difference.

Is the Map Faithful? The Deletion Test

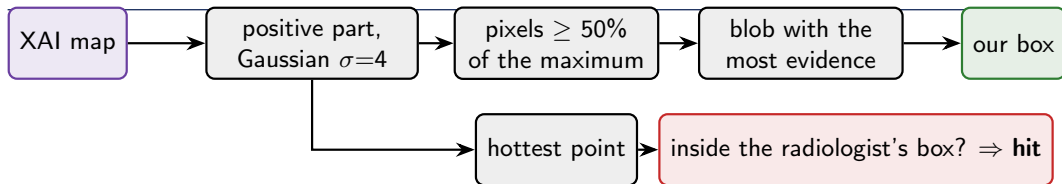
Remove the **most important** 16×16 patches first and follow the class probability: a faithful map makes it fall fast (low area). A random order is the reference.

deletion AUC	CNN	ScatNet	ResNet18
Saliency	0.530	0.573	0.663
Integrated Grad.	0.510	0.572	0.637
Guided Backprop	0.511	0.552	0.555
Grad-CAM	0.497	n/a	0.580
Occlusion	0.572	0.588	0.562
LIME	0.528	0.554	0.496
random	0.570	0.576	0.700

The winner depends on the model

- **CNN**: Grad-CAM. The last conv layer of a small CNN is where it decides.
- **ScatNet**: all close to random. Its evidence is spread over hundreds of wavelet coefficients; no small region matters alone.
- **ResNet18**: LIME. It does what the test does: removes patches.
- Saliency and IG are the least faithful on ResNet18: gradients are local and saturate in a deep net.

Does the Map Point at the Fracture?



hit rate (%)	CNN	ScatNet	ResNet18
Saliency	3.8	0.0	3.8
Integrated Grad.	2.9	2.9	23.1
Guided Backprop	3.8	0.0	4.8
Grad-CAM	0.0	n/a	9.6
Occlusion	3.8	3.8	6.7
LIME	0.0	0.0	1.0

104 fractured test X-rays; random point: 1%.

Rarely. Best: Integrated Gradients on ResNet18, **23%**.
Why so low?

- The boxes are **tiny** (1% of the image); fractures are 1–2 px at 224 px.
- A third of the fractures are **missed** by the model.
- The model also uses **shortcuts** (plates, casts), and the faithful maps honestly show them.

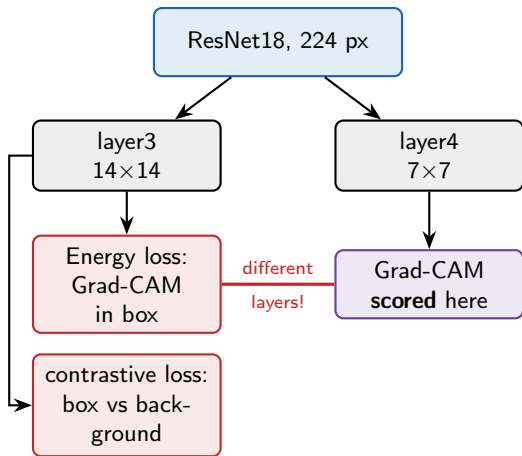
$\Rightarrow$ Faithful maps are not enough: the **model** must look at the fracture.

Part II

Teach the Model Where to Look

One network that classifies, detects, and is trained on its own explanation

A First Attempt That Failed

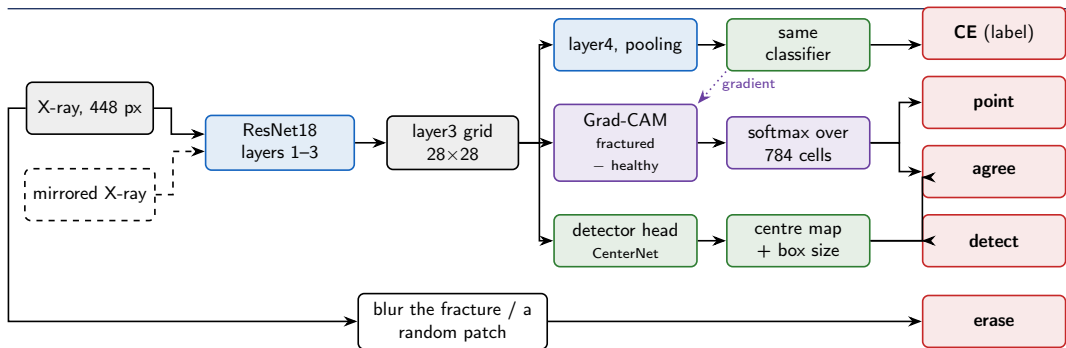


Result: Grad-CAM hit rate 10.6% $\rightarrow$ 10.6%, and **5.8%** with the contrastive loss.

Three reasons, three lessons

1. Trained on layer3, **scored on layer4** $\Rightarrow$ score the layer you train.
2. Grad-CAM's ReLU let the map **collapse to zero**: no gradient left $\Rightarrow$ use a smooth target (softmax).
3. The contrastive loss hit its **lowest value at once**: a box on bone vs black background is trivially different $\Rightarrow$ compare things that are hard to tell apart.

Our Joint Model



- **448 px**: four times the pixels, thin fractures survive.
- One **shared grid** of 16-px cells feeds three heads: classifier, detector, Grad-CAM.

- Grad-CAM is computed **during training**, on the layer that is scored.
- **A**: CE only. **B**: + detect. **C**: all five (ours).

The Five Losses: What Each Does and Why

	what it asks	why it should help
CE	the right label (batches half fractured)	the classification task
detect	CenterNet: a peak at the fracture centre + the box size	multi-task: the shared grid must learn where fractures are
point	$-\log \sum_{i \in \text{box}} p_i$, $p = \text{softmax of Grad-CAM}$	a smooth version of the hit rate; the softmax never loses its gradient
agree	$\text{JS}(p, q_{\text{detector}}) + \text{JS}(p, \text{flip}(p_{\text{mirror}}))$	SimCLR-like: two views must point at the same place; updates both heads
erase	blurred fracture $\Rightarrow$ "healthy"; blurred random patch $\Rightarrow$ still "fractured"	a map can move without the reasoning; the decision must follow the fracture

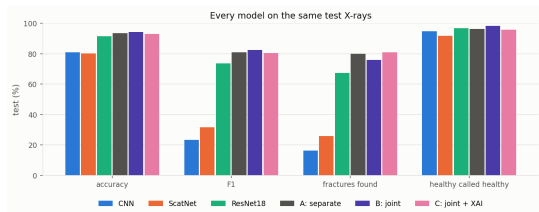
$$\mathcal{L} = \text{CE} + \lambda_d \mathcal{L}_{\text{det}} + r(t) [\lambda_p \mathcal{L}_{\text{point}} + \lambda_a \mathcal{L}_{\text{agree}} + \lambda_e \mathcal{L}_{\text{erase}}], \quad \lambda_d=1, \lambda_{p,a,e}=0.5$$

The explanation losses start after a 2-epoch warm-up $r(t)$. Same data, budget and seed for A, B and C.

A vs B vs C: Accuracy

	accuracy	F1	found	false al.	AUC
ResNet18, 224	91.5	73.7	67%	16	91.2
A: separate	93.3	81.0	80%	18	94.4
B: + detector	94.2	82.3	76%	9	94.3
C: + XAI losses	93.0	80.4	81%	21	95.2

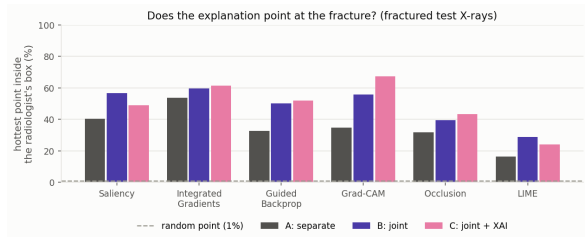
586 test X-rays; threshold chosen on val F1.



- **448 px** already helps: A finds 80% of the fractures vs 67%.
- A, B and C are **within each other's confidence intervals** (about ± 2 points).
- A second full run gave 91.1, 93.3, 93.9%: one-point gaps are noise.

Our method does not change **how often** the model is right. It changes **why**: next slides.

A vs B vs C: Where Do the Explanations Point?

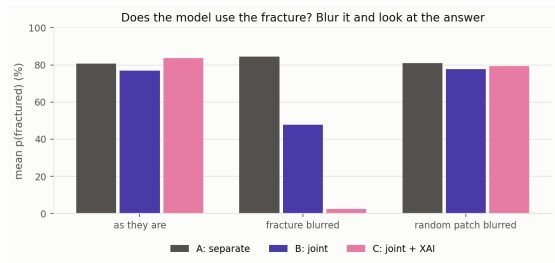


Hit rate of the six methods, 104 fractured test X-rays.

	A	C	p (A vs C)
Grad-CAM	35	67	<0.001
Guided Backprop	33	52	0.0001
Occlusion	32	43	0.050
Integrated Grad.	54	62	0.077
Saliency	40	49	0.136
LIME	16	24	0.115

- **B**: the detector alone lifts every method (mean 35 $\rightarrow$ 48%).
- **C**: Grad-CAM **doubles** (36 $\rightarrow$ 70 of 104 X-rays).
- Only Grad-CAM is in the loss, yet every method improves.

Is It Real? Blur the Fracture



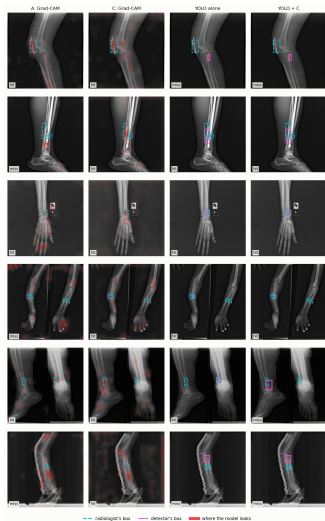
Mean $p(\text{fractured})$ of the fractured test X-rays.

	as they are	fracture blurred	random patch
A	80.6	84.4	80.9
B	77.0	47.7	77.7
C	83.7	2.5	79.4

- **A ignores the fracture:** blur it and its answer does not change.
- **C decides from the fracture:** blur it and p falls to 2.5%; blur an equal patch elsewhere and it stays at 79%.
- So the better maps are **not cosmetic**: the decision moved with them.
- Faithfulness is kept: mean deletion AUC 0.55 (A), 0.52 (B), 0.54 (C).

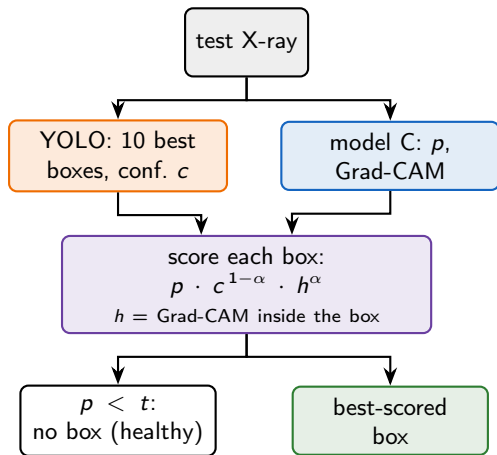
C is trained with this blur on other X-rays; the random-patch control and the five methods outside its loss are the independent evidence. Same picture in a second full run (blur drop 81 points).

Random Test X-rays (Not Picked by Hand)



- Six fractured test X-rays drawn at random.
- Columns 1–2: where **A** and **C** look (red); dashed: the radiologist's box.
- **C**'s evidence is **tighter** and sits on the bone near the break; **A**'s spreads over the hand, joints and edges.
- Columns 3–4: YOLO's box, and the box **C** chooses among YOLO's boxes (next slides).

Can Our Model Help YOLO? YOLO Proposes, C Decides

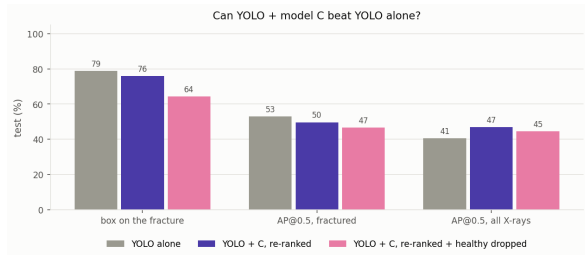


YOLOv8s (COCO-pretrained, 640 px): mAP@0.5 = **54.4%** (FracAtlas paper: 56.2%); its best box is on the fracture for **79%** of the X-rays: better than any explanation (67%).

Idea: YOLO is good at drawing boxes, C at knowing *whether* and *where*.

- α chosen on val: 0.5 (74.5% vs 71.6% for YOLO's order).
- $\alpha = 0$ is YOLO's own order.
- Option: drop the boxes of X-rays C calls healthy.

YOLO vs YOLO + C



Placing the box: no. 76% vs 79%, not significant. YOLO's ranking is already good, and C's Grad-CAM (16-px cells) is coarser than YOLO's boxes.

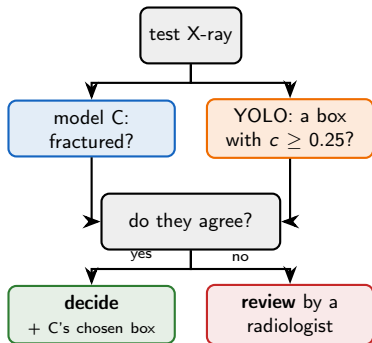
Fewer false boxes: yes. On all X-rays, AP@0.5 rises from 40.7 to **46.9**: multiplying by $p(\text{fractured})$ pushes YOLO's boxes on healthy X-rays to the bottom.

Dropping boxes is worse: C misses about 1 fracture in 5.

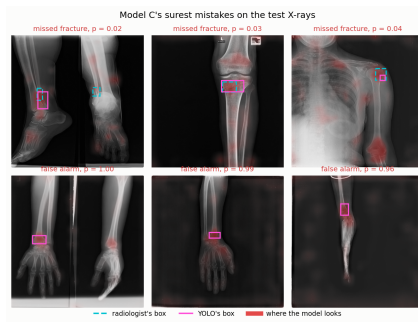
⇒ For *where*, use YOLO; for *whether*, use C.

	hit	p	AP frac.	AP all
YOLO alone	78.8	–	53.0	40.7
YOLO + C, re-ranked	76.0	0.38	49.7	46.9
+ healthy dropped	64.4	<0.001	46.6	44.6

The Pipeline: Two Opinions



decided right when it decides	76% of the X-rays
sent to review	96.6%
fractures missed	24%
	5.8% (6 of 104)



C's surest mistakes. Top: missed fractures; bottom: false alarms.

On 2 of C's 3 surest **missed** fractures YOLO's box is **on** the fracture: the pipeline asks for a review instead of missing them.

Every Method Side by Side

	what it learns	acc.	F1	best hit	good at
CNN	filters from 2,920 X-rays	80.9	23.3	4%	– (underfits)
ScatNet	only the classifier	80.0	31.6	4%	stable features, no fractures
ResNet18	fine-tunes ImageNet	91.5	73.7	23%	classification
A: 448 px	same, 4× the pixels	93.3	81.0	54%	finding fractures
B: + detector	+ where the centre is	94.2	82.3	60%	fewest false alarms
C: ours	+ where it looks, blur test	93.0	80.4	67%	looking at the fracture
YOLOv8	only boxes	–	–	79% (box)	placing the box

“best hit”: best of the six maps (for YOLO: its most confident box). Same 586 test X-rays for all; A, B, C and YOLO at 448–640 px, the others at 224 px.

Limitations and Lessons

Limitations

- One dataset; one seed per run (two full runs of A, B, C).
- 104 fractured test X-rays: 4 points = 4 X-rays. Deletion test on 10 X-rays.
- Loss weights, thresholds, α set once, not tuned.
- The pointing game is strict: one point, a tiny box.
- C is trained with the blur it is tested with (controls: random patch, five untrained maps).
- The built-in detector stays far behind YOLO (AP 22–31% vs 53%).

Lessons

- **Accuracy is not enough:** 80% can mean “finds a fifth of the fractures”.
- **Pretraining and resolution** beat hand-crafted invariance.
- A faithful map can still show a **shortcut**; the model must be **taught** to look.
- **Score the layer you train**, and test the decision, not only the map.
- A **specialist** stays best at its job: use YOLO for boxes, our model as a second opinion.

Conclusion

- **CNN vs ScatNet**: statistically tied and close to “never fractured”; a pretrained ResNet18 reaches **91.5%**.
- **Six XAI methods**: the most faithful depends on the model (Grad-CAM for the CNN, LIME for ResNet18); our Occlusion equals Captum; but the maps point at the fracture at most **23%** of the time.
- **Our joint model** (classifier + detector + explanation losses, 448 px):
 - keeps the accuracy (**93.0%**);
 - Grad-CAM hits the fracture **67%** of the time instead of 35%;
 - blurring the fracture now changes the answer (84% → **3%**).
- **YOLO** remains the best box finder (79%); our model helps it by removing false boxes (AP on all X-rays 40.7 → 46.9), and together they decide 76% of the X-rays with **97%** accuracy.

Thank you for your attention!

Code, results and figures: github.com/silvergjecha22/bone-fracture-detection